ASSIGNMENT--2

Task 1. Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.

Answer:

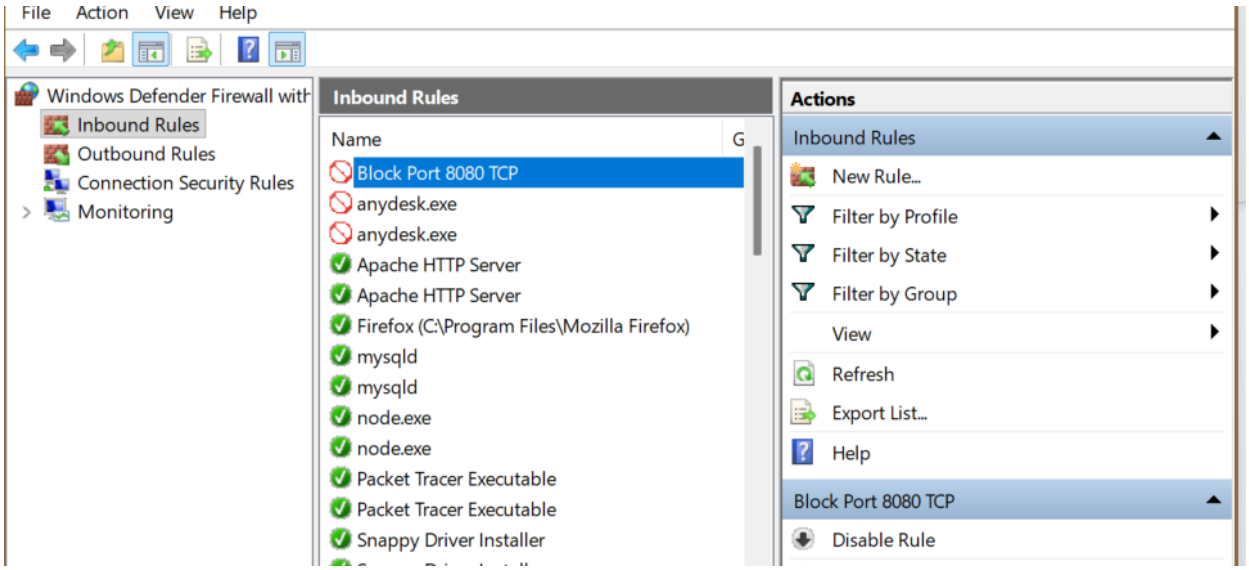
I created an inbound rule in Windows Defender Firewall with Advanced Security to block port 8080. I made it for TCP and then the same for UDP, so that all incoming traffic on this port is blocked.

1. Press Windows + R, type wf.msc and press Enter. This opens "Windows Defender Firewall with Advanced Security".
2. In the left panel click "Inbound Rules". In the right panel click "New Rule...".
3. Select "Port" and click Next.
4. Select "TCP", choose "Specific local ports", type 8080 and click Next.
5. Select "Block the connection" and click Next.
6. Keep all three profiles ticked (Domain, Private, Public) and click Next.
7. Give the rule a name, for example "Block Port 8080 TCP", add a short description and click Finish.
8. Repeat steps 2 to 7 once more, but select "UDP" in step 4 and name it "Block Port 8080 UDP".
9. The new rules now appear in the Inbound Rules list with a red "block" icon and Enabled = Yes.

The same rule can also be created from an Administrator Command Prompt:

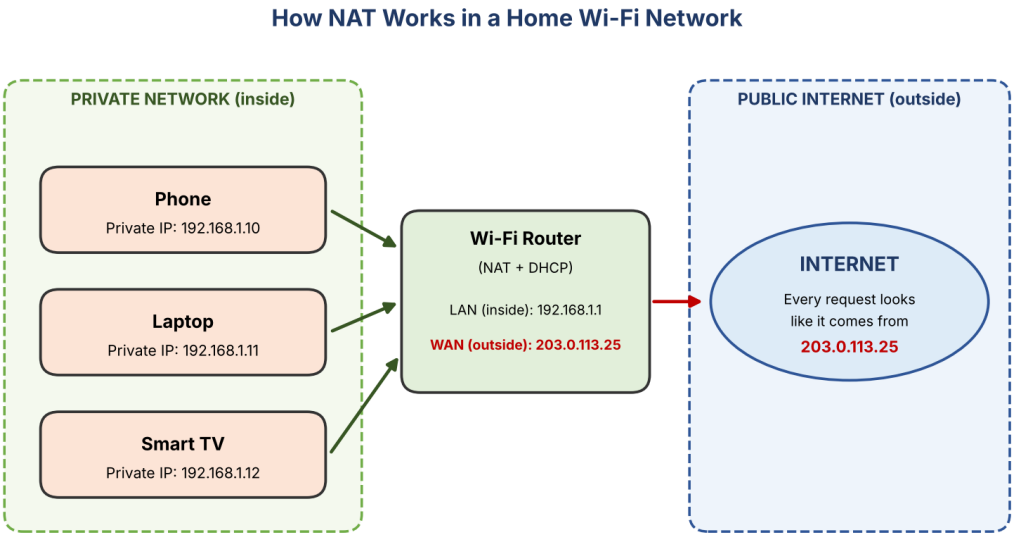
```
netsh advfirewall firewall add rule name="Block Port 8080 TCP" dir=in
action=block protocol=TCP localport=8080

netsh advfirewall firewall add rule name="Block Port 8080 UDP" dir=in
action=block protocol=UDP localport=8080
```



Task 2. Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access.

Answer:



NAT translation table (inside the router)

Private IP : Port	Public IP : Port
192.168.1.10 : 50001 (Phone)	203.0.113.25 : 40001
192.168.1.11 : 50002 (Laptop)	203.0.113.25 : 40002
192.168.1.12 : 50003 (Smart TV)	203.0.113.25 : 40003

Steps: 1) Device sends a packet with its private IP. 2) Router swaps it for the public IP and a unique port.
3) The reply returns to the public IP, and the router uses the table to deliver it to the right device.

Explanation:

- The router works as a DHCP server and gives each device a private IP address from 192.168.1.0/24: Phone 192.168.1.10, Laptop 192.168.1.11 and Smart TV 192.168.1.12. These addresses are not routable on the internet.
- The ISP gives the router only one public IP address (here 203.0.113.25, a sample address).
- When a device sends a request to the internet, the router replaces the private source IP with the public IP and a unique port number, and saves this in its NAT table.
- When the reply comes back to the public IP and that port, the router checks the NAT table and sends the data to the correct device.
- This way many devices share one public IP, and outside computers cannot directly see or reach the private devices, which also adds some security.

Task 3. Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.

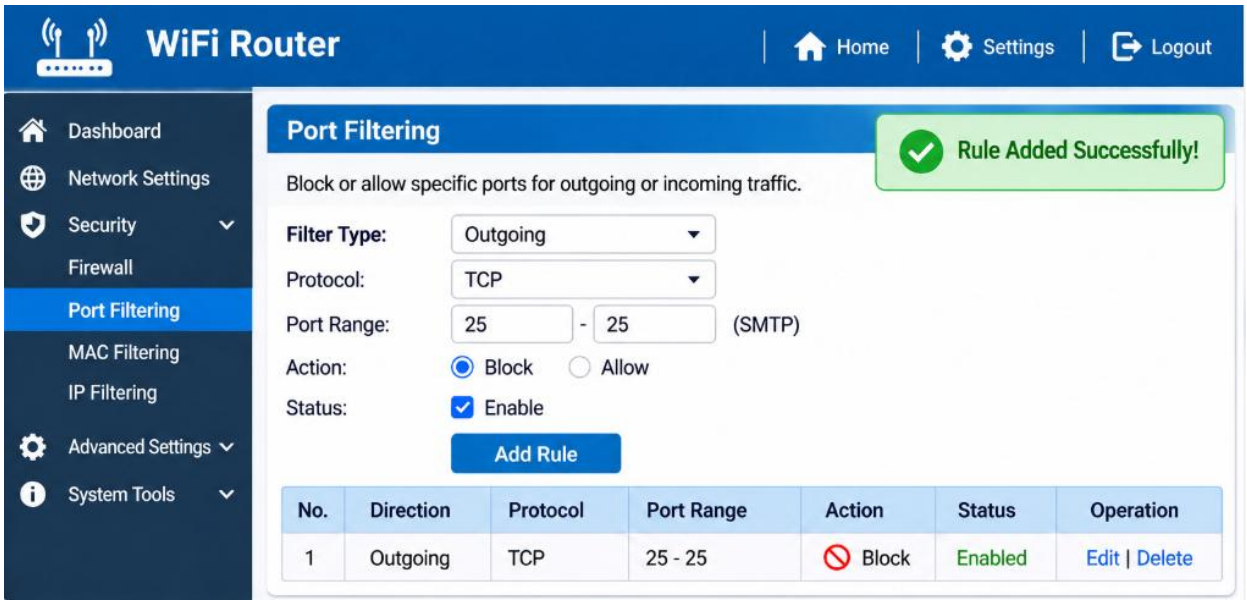
Answer:

Steps (menu names change a little from brand to brand):

1. Connect to the router and open its admin page in a browser (for example 192.168.1.1 or 192.168.0.1) and log in with the admin password.
2. Open the firewall section: Advanced / Security / Firewall, and look for "Port Filtering", "Service Filtering", "Outbound Rules" or "Access Control".
3. Click "Add" to create a new rule and enter these values: Name = Block SMTP, Direction = Outbound (LAN to WAN), Protocol = TCP, Port = 25, Action = Block / Deny.
4. Apply the rule to all devices (source IP 192.168.1.0/24, destination any).
5. Click Save / Apply, and reboot the router if it asks.

If I use a Cisco router / Packet Tracer simulator, the same thing can be done with an access list:

```
access-list 101 deny tcp any any eq 25
access-list 101 permit ip any any
interface g0/1          (the LAN-facing interface)
 ip access-group 101 in
```



How this helps stop spam (2 sentences): Port 25 is used to send email directly to other mail servers, and malware on an infected computer often uses it to send thousands of spam emails from the network. If outgoing port 25 is blocked, this spam cannot leave the network, while normal users can still send email through their mail provider on the secure submission ports 587 or 465.

Task 4. Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53.
Hint: *Focus on source/destination IPs and ports in the rule set.*

Answer:

Method: A packet filter checks the rules from top to bottom. The first rule that matches the packet's source IP, destination IP, protocol and destination port decides the action (Allow or Deny). If no rule matches, the default rule "deny all" at the end is applied.

Sample rule set used (made with AI, as the trainer allowed):

Rule	Action	Protocol	Source IP	Destination IP	Dest. port	
1	Allow	TCP	192.168.1.0/24	Any	80	HTTP
2	Allow	TCP	192.168.1.0/24	Any	443	HTTPS
3	Allow	UDP	192.168.1.0/24	Any	53	DNS
4	Deny	TCP	Any	Any	22	SSH
5	Deny	Any	Any	Any	Any	Default

The three packets, checked against the rules:

No.	Packet	Source > Destination : Port	Result	Matching rule
1	HTTP request	192.168.1.5 > 172.217.166.78 : 80 (TCP)	ALLOWED	Rule 1
2	SSH request	10.0.0.10 > 34.201.23.1 : 22 (TCP)	BLOCKED	Rule 4
3	DNS query	192.168.1.7 > 8.8.8.8 : 53 (UDP)	ALLOWED	Rule 3

Reasons:

- Packet 1 is ALLOWED: the source 192.168.1.5 is inside 192.168.1.0/24, the protocol is TCP and the destination port is 80, so Rule 1 (Allow HTTP) matches first.
- Packet 2 is BLOCKED: the source 10.0.0.10 is not in 192.168.1.0/24, so Rules 1 to 3 do not match. The destination port is 22 (SSH), so Rule 4 (Deny SSH) matches and the packet is dropped. Even if it matched nothing, the default deny rule would block it.
- Packet 3 is ALLOWED: the source 192.168.1.7 is inside 192.168.1.0/24 and the DNS query goes to port 53, so Rule 3 (Allow DNS) matches.

Note: If the trainer gives a different rule set, the same method is used: compare the source IP, destination IP and port of each packet with the rules from top to bottom.

Task 5. Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

Answer:

Instagram and Zomato do not publish their exact security setup, so these are the common, standard ways such large apps use firewalls and packet filtering.

1. Allow only the needed ports and block everything else. The servers only accept web traffic on ports 80 and 443 from the internet. Packet filtering rules deny all other ports, so risky services like SSH (22), remote desktop (3389) or databases (3306) cannot be reached from outside. This reduces the attack surface and stops port scans and break-in attempts.
2. Rate limiting and blocking bad IPs (DDoS and bot protection). The firewall and filtering layer watch how many requests come from one IP address. If an IP sends too many requests, for example thousands of login attempts on Instagram or fake order bots and menu scraping on Zomato, it is slowed down or blocked. Known malicious IP lists and fake traffic floods (DDoS) are also dropped at the edge before they reach the servers.
3. Network segmentation with a DMZ and stateful inspection. Public-facing web servers are placed in a separate zone (DMZ), and databases and internal servers (user data, payments, order data) are in a private network. The firewall only allows the web servers to talk to the internal servers on specific ports, and it allows only traffic that is part of a connection already started from inside. If an attacker enters one server, they cannot easily move to the databases.

A web application firewall (WAF) is also commonly added on top of these to block attacks such as SQL injection and cross-site scripting in the HTTP traffic.